

VIDYATRACK - QA TEST SCENARIOS

Format: Manual + Integration + Negative + Expected Failures

ROLE: PRINCIPAL

GLOBAL PRE-CONDITIONS (for all PRINCIPAL tests)

- Auth is OTP-based:
 - POST /api/v1/auth/otp/request
 - POST /api/v1/auth/otp/verify → returns TokenOut.access_token
- Authorization header: Authorization: Bearer <access_token>
- Principal test accounts:
 - P_ACTIVE: active principal
 - P_INACTIVE: inactive principal
- Principal permissions expected:
 - view_students
 - view_teachers
 - view_attendance
 - view_marks
 - parent_communication
- Principal is READ-ONLY for all academic data.

1) Feature: Principal Authentication & Dashboard Access

User Story: US-PRN-AUTH-01 Principal Login via OTP

A) HAPPY PATH (Manual/UI)

1. Enter principal phone → Request OTP
2. Enter OTP → Verify
3. Principal dashboard loads successfully

B) INTEGRATION (API)

1) Request OTP

POST /api/v1/auth/otp/request

Payload:

{ "phone": "9123456789" }

Expected:

200 { "status": "<string>" }

2) Verify OTP

POST /api/v1/auth/otp/verify

Payload:

{ "phone": "9123456789", "otp": "123456" }

Expected:

200 { "access_token": "<jwt>", "token_type": "bearer" }

3) Load principal dashboard

GET /api/v1/principal/dashboard

Expected 200 (PrincipalDashboardOut):

```
{
  "total_students": <int>,
  "total_teachers": <int>,
  "attendance_today_pct": <number>,
  "attendance_today_present": <int>,
  "attendance_today_absent": <int>,
  "attendance_today_total": <int>,
  "notices": [ ... ]
}
```

C) NEGATIVE TESTS

NEG-PRN-AUTH-01.1 Inactive principal login denied

Expected:

- 403 Forbidden or 401 Unauthorized

Must NOT:

- return access_token

NEG-PRN-AUTH-01.2 Missing/invalid token

- Access /api/v1/principal/dashboard without token

Expected:

- 401 Unauthorized

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2) Feature: Academic Oversight – View Access

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US-PRN-01: View Student List and Profiles

A) HAPPY PATH (Manual/UI)

1. Login as Principal

2. Open Students list
3. Open individual student profile

B) INTEGRATION (API)

1) List students

GET /api/v1/students

Expected 200: array of StudentOut

2) View student profile

GET /api/v1/students/{student_id}

Expected 200 (StudentProfileOut):

- personal_details
- attendance summary
- recent_results

C) NEGATIVE TESTS (P0)

NEG-PRN-01.1 Attempt to create/update/delete student

- POST /api/v1/students
- PUT/PATCH/DELETE (if attempted)

Expected:

- 403 Forbidden

NEG-PRN-01.2 Cross-school student access

- Try student_id from another school

Expected:

- 403/404

US-PRN-02: View Teachers and Assignments

A) HAPPY PATH (Manual/UI)

1. Open Teachers list
2. View teacher details and assignments

B) INTEGRATION (API)

GET /api/v1/teachers

Expected 200: array of TeacherOut

Validate:

- status (active/inactive)
- assignments visible

C) NEGATIVE TESTS (P0)

NEG-PRN-02.1 Attempt to create/update/delete teacher

- POST /api/v1/teachers
 - PATCH/DELETE teacher
- Expected:
- 403 Forbidden

US-PRN-03: View Attendance Records

A) HAPPY PATH (Manual/UI)

1. Select class/section
2. Select date
3. View attendance list/history

B) INTEGRATION (API)

GET /api/v1/attendance?date=2026-01-22

Expected 200: array of AttendanceOut

C) NEGATIVE TESTS (P0)

NEG-PRN-03.1 Attempt to create/edit attendance

- POST /api/v1/attendance
- POST /api/v1/attendance/submit

Expected:

- 403 Forbidden

US-PRN-04: View Marks and Exam Results

A) HAPPY PATH (Manual/UI)

1. Select class + subject + exam
2. View marks and remarks
3. View consolidated results

B) INTEGRATION (API)

GET /api/v1/marks?section_id=<id>&subject_id=<id>&exam_type=UNIT

Expected 200: array of MarksRecordOut

C) NEGATIVE TESTS (P0)

NEG-PRN-04.1 Attempt to record or submit marks

- POST /api/v1/marks/record
- POST /api/v1/marks/submit

Expected:

- 403 Forbidden

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3) Feature: Student Notes Visibility
User Story: US-PRN-05 View Student Notes

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A) HAPPY PATH (Manual/UI)

1. Open student profile
2. View notes section
3. Verify:
 - note text
 - author
 - timestamp

B) INTEGRATION (API)

GET /api/v1/students/{student_id}/notes
Expected 200: array of StudentNoteOut

C) NEGATIVE TESTS (P0)

NEG-PRN-05.1 Attempt to add/edit/delete notes
- POST /api/v1/students/{student_id}/notes
Expected:
- 403 Forbidden

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4) Feature: Parent Communication
User Story: US-PRN-06 Send Parent Communications

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A) HAPPY PATH (Manual/UI)

1. Compose custom message
2. Select recipients
3. Send message
4. View delivery status

B) INTEGRATION (API)

- Message send is expected to enqueue to notifications outbox
- Worker execution:
POST /api/v1/notifications/outbox/run
Expected:
200 { "processed": <int> }

C) NEGATIVE TESTS (P0)

NEG-PRN-06.1 Attempt to modify academic data from messaging flow

Expected:

- No academic endpoints invoked
- Backend blocks any attempt (403)

NEG-PRN-06.2 Messaging without parent_communication permission

Expected:

- 403 Forbidden

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5) Feature: Access Control & Read-Only Enforcement
User Story: US-PRN-SEC-01 Enforced Read-Only Access

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A) SECURITY / RBAC TESTS (P0)

For Principal role, verify ALL below return 403:

- POST /api/v1/attendance
- POST /api/v1/attendance/submit
- POST /api/v1/marks/record
- POST /api/v1/marks/submit
- POST /api/v1/students
- POST /api/v1/teachers
- POST /api/v1/teaching-assignments
- PATCH/DELETE any academic entity

B) UI VALIDATION (Manual)

- No edit/create/delete buttons visible
- All views are read-only

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6) Reliability & Audit Expectations

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- Principal views must be non-mutating (GET-only)
- No side effects from dashboard loads
- Access attempts beyond scope must be logged
- Parent messages must not expose academic data unless explicitly intended

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DEFINITION OF DONE - PRINCIPAL ROLE

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- Principal can log in and see dashboard
- All academic data is viewable but immutable
- RBAC enforced at backend for every write attempt
- Parent communication works without data mutation

- No cross-school data exposure
- No P0/P1 defects open

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END - ROLE: PRINCIPAL QA TEST SCENARIOS
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